

LYRA: GENERATIVE 3D SCENE RECONSTRUCTION VIA VIDEO DIFFUSION MODEL SELF-DISTILLATION

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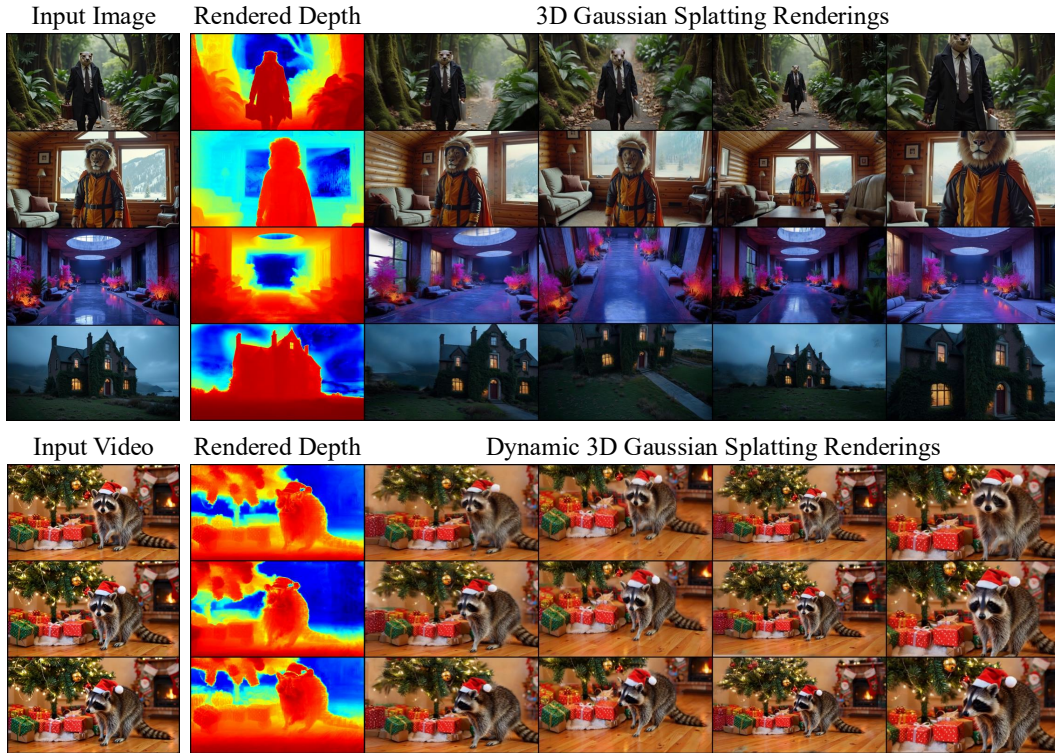


Figure 1: **Feed-Forward 3D and 4D Scene Generation.** From a single image (top), *Lyra* infers a 3D Gaussian Splatting (3DGS) representation in a feed-forward manner, by self-distilling a video diffusion model without recourse to real-world multi-view data. Given a video input (bottom), *Lyra* instead infers a dynamic 3DGS that affords interactive control over both time (rows) and viewpoint (columns).

ABSTRACT

The capacity to synthesize virtual environments constitutes a foundational prerequisite for applications spanning interactive entertainment through to the physical-AI domains of robotics, autonomous driving, and industrial automation. Prevailing learning-based approaches to 3D reconstruction remain predicated upon the availability of captured real-world multi-view observations, a resource that is seldom readily procurable. Although recent progress in video diffusion models has evidenced remarkable generative imagination, the intrinsically two-dimensional nature of their outputs circumscribes their utility for simulation settings in which an agent must navigate and interact with its surroundings. In this work, we advance a self-distillation framework whose objective is to transfer the 3D knowledge implicitly encoded within video diffusion models into an explicit 3D Gaussian Splatting (3DGS) representation, thereby obviating the requirement for multi-view training data. Concretely, we augment the conventional RGB decoder with a complementary 3DGS decoder that is supervised directly by the output of the former. Under this arrangement, the 3DGS decoder can be trained exclusively from synthetic

data generated by video diffusion models. At inference, our model synthesizes 3D scenes amenable to real-time rendering from either a textual prompt or a single image, and the framework generalizes further to the generation of dynamic 3D scenes from monocular input video. Empirical evaluation demonstrates that our framework attains state-of-the-art performance across both static and dynamic 3D scene generation.

1 INTRODUCTION

The synthesis of high-quality 3D environments at scale constitutes a long-standing challenge in computer vision and graphics, underpinning applications as diverse as film-making, VR/AR, and physical AI endowed with closed-loop simulation. Each of these applications presupposes explicit 3D representations that concurrently accommodate real-time rendering, physical interaction, and consistent multi-view synthesis.

Recent advances in neural 3D reconstruction (Mildenhall et al., 2020; Kerbl et al., 2023; Moenne-Loccoz et al., 2024) have rendered the recovery of such representations from posed imagery feasible. Their dependence upon accurate camera poses and high-quality imagery, however, severely curtails their scalability; the difficulty is compounded still further for dynamic scenes, which ordinarily necessitate synchronized multi-camera rigs (Li et al., 2022). Reconstruction methods are, moreover, inherently confined to the observed content and are consequently incapable of extrapolating beyond the input views. Although a body of recent work seeks to attenuate these limitations through feed-forward reconstruction models (Zhang et al., 2024e; Ren et al., 2024c), such approaches encounter a bottleneck of their own—namely, the scarcity of diverse, large-scale 3D training data, a deficiency that engenders poor out-of-domain generalization.

Video diffusion models (Cosmos, 2025; Wan, 2025) constitute a compelling alternative. Having been trained on massive internet-scale video corpora, they attain impressive fidelity and generalization across heterogeneous environments. By assimilating real-world videos exhibiting varied camera trajectories—ranging from handheld motion through cinematic panning to aerial drone footage—these models implicitly encode cues concerning the underlying 3D world without recourse to multi-view training data. In contradistinction to reconstruction approaches (Charatan et al., 2024), generative models are further capable of hallucinating plausible content beyond that which is visible in the input frames. Nonetheless, video diffusion models yield only 2D frames and thus lack any explicit 3D representation—a shortcoming that restricts their applicability to simulation and to downstream tasks predicated on geometric consistency, long-term coherence, and physical interaction. Encouragingly, recent work (Wang et al., 2024e; Ren et al., 2025) has demonstrated that video models can be adapted for explicit camera control by leveraging comparatively small-scale posed datasets. This capacity to generate posed image sequences transforms them into a potent instrument that may serve simultaneously as input to, and supervision for, 3D reconstruction models.

Motivated by this insight, we reconcile these two hitherto disjoint paradigms—reconstruction and generation—within a formulation we term *generative 3D scene reconstruction*. Specifically, we introduce *Lyra*, a novel method that generates explicit 3D environments from the latent representations of video diffusion models within a single forward pass. Crucially, *Lyra* is trained under a self-distillation framework (Fig. 2) in which a 3D Gaussian Splatting (3DGS) decoder operates directly within the latent space of a video diffusion model. Given a single input image or video, we sample a camera trajectory as conditioning input, denoise the resulting video latent, and decode it along two parallel branches: the first employs the standard RGB decoder to synthesize a video sequence, whereas the second—our 3DGS decoder—yields an explicit 3D representation. Taken together, these branches constitute a self-distillation loop wherein the RGB branch (teacher) supervises the 3DGS branch (student). To broaden viewpoint coverage, we further sample multiple camera trajectories, generate the corresponding video latents, and train the 3DGS decoder to fuse the information distributed across them while simultaneously attenuating long-term and multi-view inconsistencies.

This generative reconstruction framework confers three principal benefits. **(i)** It enables the generation of large-scale synthetic environments spanning heterogeneous scenarios directly from video diffusion models, thereby dispensing with real-world multi-view captures altogether. **(ii)** By virtue of operating within the latent space of a video model, it affords the efficient processing of numerous views while circumventing the substantial memory overhead incurred by pixel-space feed-forward reconstruction methods. **(iii)** Its explicit 3DGS output guarantees geometric consistency, furnishing representations

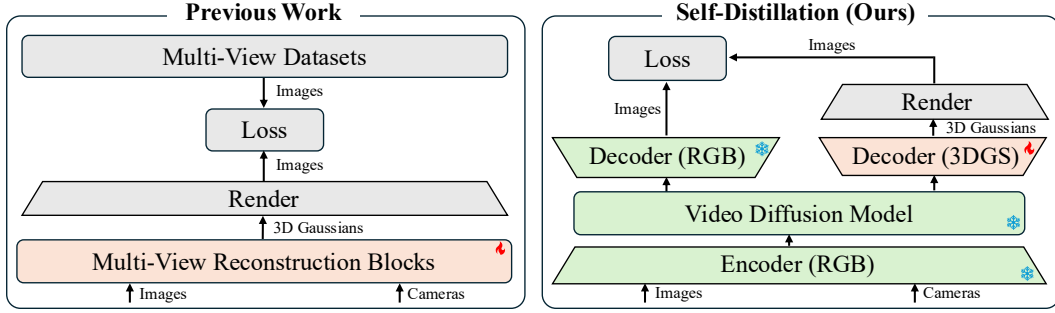


Figure 2: **Self-distillation framework of Lyra.** Prior work (left) (Szymanowicz et al., 2025b) trains multi-view reconstruction blocks upon real-world datasets of limited diversity (Zhou et al., 2018; Ling et al., 2024b). In contradistinction, we advance a self-distillation framework (right) for generative 3D scene reconstruction, wherein a pre-trained camera-controlled video diffusion model, through its RGB decoder output (teacher), supervises the rendering of the 3DGS decoder (student). Recourse to a video model pre-trained on diverse 2D video data enables us to furnish correspondingly diverse multi-view supervision.

that are immediately amenable to downstream tasks such as physical simulation. Consequently, at inference time, Lyra generates high-quality 3D Gaussian scenes from monocular input that admit real-time rendering, *without recourse to any additional optimization or post-processing*.

We additionally extend this self-distillation framework to dynamic 4D generation from monocular video. In this regime, the video model (teacher) furnishes space–time supervision, while the student is trained to produce time-conditioned 3DGS representations that enable novel-view synthesis of dynamic scenes.

In sum, our work advances the following contributions:

- We introduce a self-distillation framework that trains a 3DGS student decoder using a pre-trained camera-controlled video diffusion model as its teacher, thereby eliminating any dependence upon captured multi-view real-world data.
- We demonstrate that the framework extends naturally to dynamic scenes, thereby supporting 4D reconstruction from a single monocular video.
- We show that our model generalizes across markedly diverse scenes, attaining state-of-the-art results in both single-image 3D scene generation and single-video 4D scene generation.

2 RELATED WORK

Multi-view image generation. Early efforts in multi-view image generation (Watson et al., 2023; Liu et al., 2023; Shi et al., 2024; Wang & Shi, 2023) were directed predominantly at object-centric scenes devoid of background, whereas subsequent contributions (Sargent et al., 2024; Tang et al., 2023b; Schneider et al., 2025) extended such generators to the scale of full scenes. CAT3D (Gao et al., 2024b), for instance, repurposes a pre-trained image diffusion model for multi-view image generation from an arbitrary generated or real image input; in a subsequent stage, the resulting multi-view images are reconstructed into an explicit 3D representation such as a NeRF (Mildenhall et al., 2020) or 3DGS (Kerbl et al., 2023). This paradigm has, moreover, been generalized to the synthesis of dynamic 3D scenes from multiple viewpoints by means of a space-time diffusion model (Watson et al., 2024; Wu et al., 2025b; Kuang et al., 2024; Wang et al., 2025a). Notwithstanding the impressive results these methods achieve, grounding their generations in an explicit 3D representation entails an expensive optimization stage whose cost is not amortized across scenes. Rather than first synthesizing multi-view images and thereafter reconstructing them through optimization, we pursue the *direct* generation of a 3D scene from text or a single image.

Camera-conditioned video generation. Considerable progress has been registered in fine-tuning video diffusion models for 3D camera control. MotionCtrl (Wang et al., 2024e) pioneered such control by conditioning pre-trained video models on camera poses, whereupon follow-up works (He et al., 2024a; Xu et al., 2024c; Bahmani et al., 2025a;b) came to represent cameras as Plücker coordinates so as to afford pixel-wise conditioning. A distinct line of inquiry (Hu et al., 2024; Hou et al., 2024; Zhang et al., 2024b) instead investigates training-free camera control of video diffusion models. More recently still, SynCamMaster (Bai et al., 2025b) and ReCamMaster (Bai et al., 2025a)

have demonstrated impressive camera-controlled video synthesis predicated on large-scale synthetic training data. Although these works attain compelling results, their outputs remain ungrounded in 3D. Our contribution is orthogonal to this line of research, inasmuch as we address the task of feed-forward 3D generation from camera-controlled video models, thereby permitting the rendering of novel viewpoints *after* generation. Concretely, we build upon GEN3C (Ren et al., 2025), a recent camera-controlled video diffusion model, which we adopt as the teacher within a self-distillation framework for 3D reconstruction.

Feed-forward 3D models. Early work on feed-forward 3D reconstruction was concerned principally with object-centric scenes lacking any background, predicting a NeRF (Hong et al., 2024; Li et al., 2024b) or 3DGS (Tang et al., 2024a; Szymanowicz et al., 2024) directly from either text or an image. More recent contributions (Charatan et al., 2024; Zhang et al., 2024e; Szymanowicz et al., 2025a; Ren et al., 2024c) shift their focus to scene-level 3D reconstruction, characteristically regressing per-pixel 3D Gaussians from one or several input images. Most scene-level reconstruction methods, however, remain confined to scenes drawn from the training distribution—such as RealEstate10K (Zhou et al., 2018)—and consequently exhibit limited generalizability to generated scenes. A further body of recent work extends feed-forward 3D reconstruction models toward the generation of scenes. Bolt3D (Szymanowicz et al., 2025b), for its part, trains a pointmap (Wang et al., 2024b) autoencoder to produce pointmaps alongside the multi-view images, which are subsequently exploited for feed-forward 3D reconstruction. Most closely related to our own approach, Wonderland (Liang et al., 2025a) employs a camera-controlled video model to generate 3D Gaussians by way of a feed-forward network. In contradistinction to these works, we deploy a camera-controlled video model as a teacher within a self-distillation framework so as to train our student 3D model *without recourse to real-world multi-view data*. Of particular note, we furthermore demonstrate a straightforward extension of the framework to feed-forward 4D scene generation, a task hitherto unexplored.

3 SELF-DISTILLATION USING VIDEO DIFFUSION MODELS

The central idea underpinning our approach is to distill the 3D knowledge implicitly embedded within video diffusion models into an explicit 3DGS decoder capable of yielding high-quality 3D representations. To this end, we construct a teacher–student framework in which the video diffusion model (teacher) generates RGB videos that in turn supervise the 3DGS decoder (student) operating within the selfsame latent space, as illustrated in Fig. 2. Throughout this section we first elaborate our video diffusion model backbone (Sec. 3.1) together with its role in the self-distillation of the 3DGS decoder (Sec. 3.2). We subsequently motivate the design choices governing the 3DGS decoder in Sec. 4 and delineate the minimal modifications requisite for adapting the pipeline to dynamic 3D scenes in Sec. 5.

3.1 BACKGROUND: CAMERA-CONTROLLED AND 3D-CONSISTENT VIDEO DIFFUSION

Because our 3DGS decoder is supervised exclusively by the camera-controlled video diffusion model, we depend critically upon the latter’s 3D consistency. For this reason, we ground our approach in GEN3C (Ren et al., 2025), a recent camera-conditioned video diffusion model, whose salient design choices we survey in what follows.

Spatiotemporal 3D cache. With a view to enhancing video consistency and the precision of camera control, GEN3C constructs a spatiotemporal 3D cache $\{\mathbf{P}^{t,v}\}$ from the input image(s) or videos, wherein each $\mathbf{P}^{t,v}$ denotes a colored point cloud obtained by unprojecting the depth estimate (Wang et al., 2024a) of an RGB image captured from camera viewpoint v at time t . The cache is organized as an $L \times V$ array, in which L designates the number of frames (i.e., the temporal length) and V the number of camera views.

Rendering and structured guidance. In order to exploit this cache, GEN3C renders each point cloud from arbitrary prescribed camera poses, thereby producing RGB images $\mathbf{I}^{t,v}$ and disocclusion masks $\mathbf{M}^{t,v}$ via $(\mathbf{I}^{t,v}, \mathbf{M}^{t,v}) = \mathcal{R}(\mathbf{P}^{t,v}, \mathbf{C}^t)$, where $\mathcal{R}$ denotes the rendering function that projects the 3D point cloud $\mathbf{P}^{t,v}$ onto the 2D camera plane in accordance with the camera pose $\mathbf{C}^t$ at time t . Given a sequence of camera poses $\mathbf{C} = (\mathbf{C}^1, \dots, \mathbf{C}^L)$, rendering the entirety of the cache elements yields V videos of length L , which may be stacked into image sequences $\mathbf{I}^v \in \mathbb{R}^{L \times 3 \times H \times W}$ and mask sequences $\mathbf{M}^v \in \mathbb{R}^{L \times 1 \times H \times W}$ for each view v . The disocclusion masks demarcate those regions that

the video diffusion model is tasked with completing, and the resulting renderings serve, collectively, as structured visual guidance for the subsequent video generation.

Video variational autoencoder (VAE). In diffusion-based video generation models, a video variational autoencoder (VAE) (Kingma & Welling, 2013; Rombach et al., 2022) is customarily enlisted to compress videos into a lower-dimensional latent space, so as to render both training and inference tractable. Given an RGB video $\mathbf{I} \in \mathbb{R}^{L \times 3 \times H \times W}$, a pre-trained VAE encoder $\mathcal{E}$ maps the video into a latent space, i.e., $\mathbf{z} = \mathcal{E}(\mathbf{I}) \in \mathbb{R}^{L' \times C \times h \times w}$, whereupon the training and inference of the diffusion model are conducted entirely within this latent space. The final video $\hat{\mathbf{I}} = \mathcal{D}_{rgb}(\mathbf{z})$ is thereafter recovered by a pre-trained VAE decoder $\mathcal{D}_{rgb}$. Throughout this paper we adopt the pre-trained GEN3C (Ren et al., 2025; Cosmos, 2025) model, for which the latent channel dimension is $C = 16$, the temporal dimension is $L' = (L - 1)/\tau + 1$, and the spatial dimensions are $h = H/\sigma$ and $w = W/\sigma$, the temporal and spatial compression factors being $\tau = 8$ and $\sigma = 8$, respectively.

3.2 SELF-DISTILLATION

We train the 3DGS decoder $\mathcal{D}_s$ as a student under a teacher–student paradigm in which a camera-controlled video diffusion model $\mathcal{V}$ assumes the role of teacher. To secure diverse supervision, we curate a large-scale collection of heterogeneous text prompts by means of large language models (OpenAI, 2025; Bai et al., 2023), synthesize images I with an image diffusion model (Black Forest Labs, 2024), and thereafter expand these into multi-view sequences via GEN3C (Ren et al., 2025).

Teacher–student setup. Given an input image I together with a sampled camera trajectory $\{\mathbf{C}^t\}_{t=1}^L$, the video diffusion model $\mathcal{V}$ generates a denoised video latent $\mathbf{z} = \mathcal{V}(I, \{\mathbf{C}^t\}_{t=1}^L)$. This latent $\mathbf{z}$ is then decoded along two branches: the pre-trained RGB decoder $\mathcal{D}_{rgb}$ produces video frames $\mathbf{I}_{\mathcal{D}_{rgb}} = \mathcal{D}_{rgb}(\mathbf{z})$, whereas the 3DGS decoder $\mathcal{D}_s$ emits explicit 3D Gaussians $\mathbf{G}$. The views rendered from the 3DGS (*student*), defined as $\mathbf{I}_{\mathcal{D}_s} = \text{Render}(\mathbf{G}, \{\mathbf{C}^t\}_{t=1}^L)$, are supervised so as to accord with the RGB frames $\mathbf{I}_{\mathcal{D}_{rgb}}$ (*teacher*), thereby closing the self-distillation loop.

Multi-trajectory supervision. In order to enlarge the viewpoint coverage afforded by a single input image, we sample $V = 6$ camera trajectories per image, as depicted in Fig. 3. For each trajectory v we construct a spatiotemporal cache $\{\mathbf{P}^{t,v}\}_{t=1}^L$ together with $L = 121$ camera poses $\{\mathbf{C}^{t,v}\}_{t=1}^L$. Propagating these caches through the video diffusion model $\mathcal{V}$ yields latents $\mathbf{z}^v$, which the teacher decodes into RGB frames $\mathbf{I}_{\mathcal{D}_{rgb}}^v = \mathcal{D}_{rgb}(\mathbf{z}^v)$. The 3DGS decoder $\mathcal{D}_s$ thereupon learns to fuse these multiple latents $\mathbf{z}^v$ into coherent Gaussians $\mathbf{G}$ while simultaneously in-painting the disoccluded regions. By virtue of this self-distillation design, the 3DGS decoder $\mathcal{D}_s$ is trained wholly from the synthetic supervision furnished by $\mathcal{V}$. Operating within latent space permits the efficient aggregation of numerous trajectories, whereas the explicit Gaussian output $\mathbf{G}$ enforces geometric consistency and thereby underwrites downstream simulation and real-time rendering.

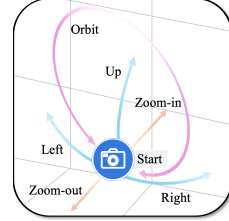


Figure 3: Six camera trajectories sampled so as to maximize view coverage.

4 FEED-FORWARD RECONSTRUCTION FROM MULTI-VIEW VIDEO LATENTS

Our feed-forward 3DGS decoder $\mathcal{D}_s$ is devised to transform the synthesized multi-view latents produced by our video model $\mathcal{V}$ into an explicit 3D representation that may be rendered from arbitrary viewpoints. We elect 3D Gaussian Splatting (3DGS) (Kerbl et al., 2023) on account of both its explicit representation and its expeditious rendering speed, and we furnish a detailed visualization of the decoder in Fig. 4 (right).

4.1 3DGS DECODER

Scaling Latent-Based 3D Reconstruction. Prior feed-forward reconstruction frameworks generate 3D Gaussians on a per-pixel basis and are, in consequence, constrained by both the number and the resolution of the input views they can accommodate. GS-LRM (Zhang et al., 2024e), by way of illustration, operates on merely 2–4 images at 512×512 resolution, whereas AnySplat (Jiang et al., 2025a) is trained with 24 images at 448×448 resolution. Our video model, by contrast, synthesizes $V \times L = 6 \times 121 = 726$ input views at 704×1280 resolution—a scale that lies far beyond the capacity of existing methods and that would exceed the available GPU memory. The

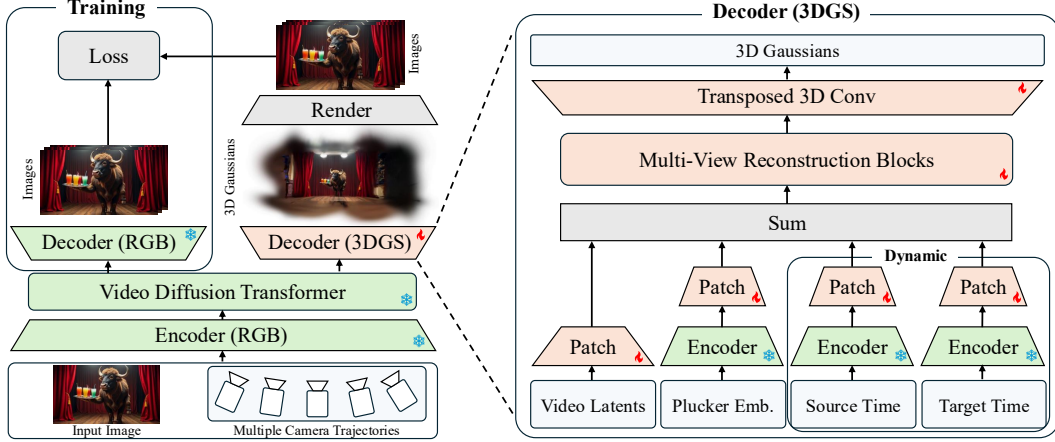


Figure 4: **3D Generative Reconstruction Framework.** Our pipeline is founded upon a camera-controlled video diffusion model (Ren et al., 2025) pre-trained on large-scale data. We train a 3D Gaussian Splatting (3DGS) decoder by aligning the 2D image renderings of the generated 3DGS scenes with the RGB-decoded generations of the pre-trained video model, training the 3DGS decoder alone while the pre-trained autoencoder and diffusion model remain frozen. At inference time we invoke the 3DGS decoder directly, the RGB decoder being no longer required. Time conditioning within the 3DGS decoder, moreover, permits us to extend the approach effortlessly from static to dynamic 3D scene generation.

bottleneck resides in the attention mechanism applied to the visual tokens, whose memory and compute requirements escalate with the number of pixels. To surmount this limitation, we eschew scaling in pixel space and instead operate directly within the compressed video latent space furnished by our camera-controlled video diffusion model. The resulting multi-view video latents are denoted $\mathbf{Z} \in \mathbb{R}^{V \times L' \times C \times h \times w}$.

Architecture. The 3DGS decoder $\mathcal{D}_s$ first projects its inputs into the hidden dimension of the principal reconstruction blocks and thereafter emits per-pixel 3D Gaussian features $\mathbf{G} \in \mathbb{R}^{V \times L \times H \times W \times 14}$. The inputs in question comprise the multi-view video latents $\mathbf{Z}$ and the encoded Plücker embeddings $\mathbf{E}$, both possessing latent dimensions $\mathbf{Z}, \mathbf{E} \in \mathbb{R}^{V \times L' \times C \times h \times w}$. Each input component is first patchified so as to conform to the hidden dimension of the principal reconstruction blocks. Notably, we dispense with any auxiliary visual encoder, requiring only a spatial 2×2 patchification layer (Dosovitskiy et al., 2021) to render the video latents into flattened tokens for the reconstruction network. The sum of the two inputs is subsequently propagated through the reconstruction network. We adhere to the design introduced in Long-LRM (Ziwen et al., 2024), whereby a single block consists of one Transformer (Vaswani et al., 2017) layer followed by seven Mamba-2 (Dao & Gu, 2024) layers; repeating this block twice furnishes 16 layers with 512 hidden dimensions. Finally, a transposed 3D convolution maps the hidden representation onto 14 Gaussian channels—namely, the 3D position (x, y, z) , scale (s_x, s_y, s_z) , rotation quaternion (q_w, q_x, q_y, q_z) , opacity α , and RGB (r, g, b) . Formally, the static decoder is expressed as $\mathbf{G} = \mathcal{D}_s(\mathbf{Z}, \mathbf{E})$.

Plücker embeddings. Raw Plücker embeddings are first derived from the camera poses $\{\mathbf{C}^{t,v}\}$ and intrinsics as $\mathbf{E}^{\text{raw}} \in \mathbb{R}^{V \times L \times 6 \times H \times W}$. To encode these embeddings we repurpose the RGB encoder $\mathcal{E}$ of the pre-trained VAE: concretely, we separately encode the 3-dimensional ray directions and the 3-dimensional cross product of the ray directions and origins by means of $\mathcal{E}$, and concatenate the resulting latents along the channel dimension to obtain $\mathbf{E}^{\text{enc}} \in \mathbb{R}^{V \times L' \times 2C \times h \times w}$. A lightweight MLP thereafter maps $\mathbf{E}^{\text{enc}}$ to $\mathbf{E} \in \mathbb{R}^{V \times L' \times C \times h \times w}$, halving the channel dimension so as to match that of the video latent $\mathbf{Z}$.

4.2 LOSS FUNCTION

Image-based supervision. We supervise our 3DGS decoder with an image-based reconstruction loss that decomposes into a Mean Squared Error (MSE) term $\mathcal{L}_{mse}$ and an LPIPS term $\mathcal{L}_{lips}$ (Zhang et al., 2018), the latter employing VGG (Simonyan & Zisserman, 2014) as its feature extractor.

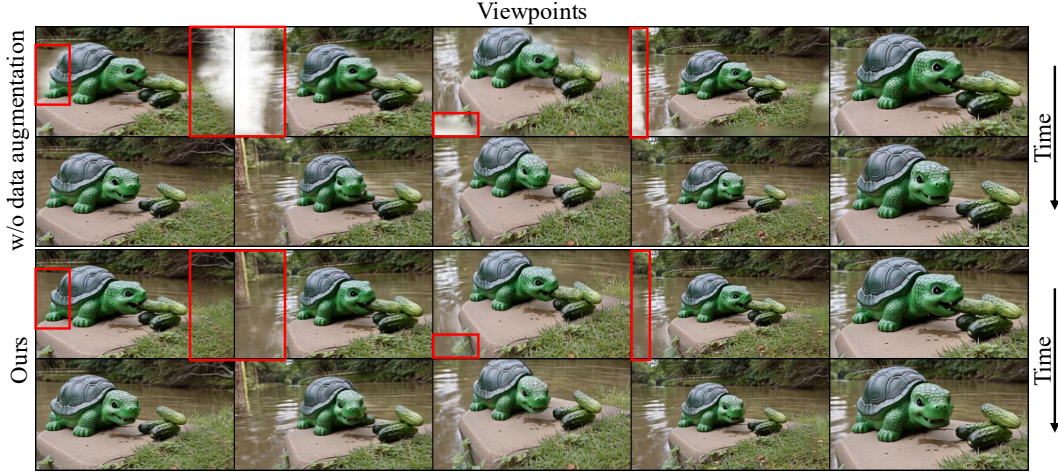


Figure 5: **Dynamic data augmentation.** Naive supervision of the time-aware 3DGS decoder gives rise to artifacts in the generated 3D Gaussians; in particular, early timesteps viewed from extreme poses exhibit low opacity throughout the regions left uncovered by the supervision signal. To remedy this, we augment the supervision data with a motion-reversed video, thereby ensuring that each timestep is observed under full spatial coverage and, in turn, obviating the low-opacity artifacts at early timesteps.

Depth supervision. We observed that a decoder trained under the RGB loss alone tends to yield flattened geometry. To remedy this pathology, we additionally supervise the rendered depth maps against the consistent video depth estimated by the off-the-shelf system ViPE (Huang et al., 2025), invoking the scale-invariant depth loss $\mathcal{L}_{depth}$ of Long-LRM (Ziwen et al., 2024).

Opacity-based pruning. In the manner of Long-LRM (Ziwen et al., 2024), we impose an L_1 regularization on the opacity, $\mathcal{L}_{opacity}$, and excise the Gaussians occupying the lowest 80% of opacity.

Total loss. The total loss is constituted as the weighted sum of the foregoing terms, with weighting factors λ_i ,

$$\mathcal{L} = \lambda_{mse} \mathcal{L}_{mse} + \lambda_{lpips} \mathcal{L}_{lpips} + \lambda_{depth} \mathcal{L}_{depth} + \lambda_{opacity} \mathcal{L}_{opacity} \quad (1)$$

where we set $\lambda_{mse} = 1.0$, $\lambda_{lpips} = 0.5$, $\lambda_{depth} = 0.05$, and $\lambda_{opacity} = 0.1$.

5 EXTENSION TO DYNAMIC 3D SCENES

Our approach admits extension to dynamic 3D scenes at the expense of only minimal modifications, which we outline below.

Self-distillation for dynamic scenes. Our dynamic 3D configuration hews closely to the design of its static counterpart. In lieu of a single image, the video model now ingests a single video of length L together with the corresponding camera poses, and generates multi-view video latents that capture the same underlying motion. For video inputs we adopt a protocol analogous to the static case, sampling diverse text prompts with large language models and thereafter generating videos with video diffusion models (Cosmos, 2025; Wan, 2025). The generated videos are subsequently annotated with camera poses and depth maps by means of ViPE (Huang et al., 2025).

Dynamic 3DGS decoder architecture. Following the bullet-time design of Liang et al. (2025b), we contrive our dynamic 3DGS decoder $\mathcal{D}_s$ to generate time-dependent 3D Gaussians for the totality of the video frames. The dynamic decoder $\mathcal{D}_s$ mirrors the architecture of $\mathcal{D}_s$ closely, save that we augment its input with the encoded source and target time embeddings $\mathbf{T}^{src}, \mathbf{T}^{tgt} \in \mathbb{R}^{V \times L' \times C \times h \times w}$. The unencoded (raw) source times are assigned to each input frame, yielding $\mathbf{T}^{src, raw} \in \mathbb{R}^{V \times L \times 1 \times H \times W}$, while the target time is $\mathbf{T}^{tgt, raw} \in \mathbb{R}^{V \times 1 \times 1 \times H \times W}$. Both are spatially replicated and augmented with a 2-dimensional sinusoidal embedding so as to expand the channel dimension to 3, whereupon they are encoded with the RGB encoder $\mathcal{E}$ to latent dimensions $\mathbf{T}^{src}, \mathbf{T}^{tgt} \in \mathbb{R}^{V \times L' \times C \times h \times w}$. The target time is replicated along the latent temporal dimension so as to match

Table 1: **State-of-the-art comparisons.** We benchmark our method against prior works on single-image-to-3D generation across the RealEstate10K, DL3DV, and Tanks-and-Temples datasets.

Method	RealEstate10K			DL3DV			Tanks-and-Temples		
	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$
ZeroNVS	13.01	0.378	0.448	13.35	0.339	0.465	12.94	0.325	0.470
ViewCrafter	16.84	0.514	0.341	15.53	0.525	0.352	14.93	0.483	0.384
Wonderland	17.15	0.550	0.292	16.64	0.574	0.325	15.90	0.510	0.344
Bolt3D	21.54	0.747	0.234	-	-	-	-	-	-
Ours	21.79	0.752	0.219	20.09	0.583	0.313	19.24	0.570	0.336

L' . During training, a target time step is sampled at random, and the corresponding 3D Gaussians are decoded and supervised across all trajectories generated under dynamic data augmentation. We fine-tune $\mathcal{D}_d$ from a pre-trained $\mathcal{D}_s$, initializing the patchification layers of $\mathbf{T}^{\text{src}}$ and $\mathbf{T}^{\text{tgt}}$ to zero; $\mathbf{T}^{\text{src}}$ and $\mathbf{T}^{\text{tgt}}$ are then added directly to the sum of $\mathbf{Z}$ and $\mathbf{E}$. Formally, the dynamic decoder is defined as $\mathbf{G} = \mathcal{D}_d(\mathbf{Z}, \mathbf{E}, \mathbf{T}^{\text{src}}, \mathbf{T}^{\text{tgt}})$.

Dynamic data augmentation. In static 3D scenes, where time does not enter as a factor, frames drawn from any timestep may serve to supervise the 3DGS renderings. For dynamic 3D scenes, by contrast, the 3DGS evolves over time, so that only frames bearing the corresponding timestamp constitute valid supervision. This restriction is liable to induce a degenerate solution, in which the 3DGS at a given timestep simply disregards the information conveyed by frames at other timesteps (see Fig. 5).

To forestall this pathology, we introduce a dynamic data augmentation strategy that equilibrates supervision across timesteps. Early frames are, by their nature, associated with viewpoints proximate to the input image, whereas later frames correspond to more distant viewpoints; to counteract this imbalance, we augment the training data with paired supervision views for each timestep—one near and one far. Concretely, we reverse the input video in frame order and submit it to the video model $\mathcal{V}$, which produces six supplementary multi-view sequences. Reversing these sequences back to the original motion order yields six trajectories that progress inward—from distant viewpoints toward the input—and, in conjunction with the original six outward trajectories, furnishes 12 supervision views per timestep. Crucially, this augmentation is confined to training alone: the reversed trajectories operate purely as auxiliary supervision signals that avert collapse arising from pruning artifacts, and are dispensable at inference time.

6 EXPERIMENTS

6.1 EXPERIMENTAL SETUP

Datasets. To train our model we forgo any existing multi-view dataset, electing instead to construct our own—which we designate the *Lyra* dataset—and relying solely upon our video model to supervise the 3DGS decoder. The 3D reconstruction setup draws upon 59,031 images, whereas the 4D setup comprises 7,378 videos. All data derive from diverse text prompts and span scenarios as varied as indoor and outdoor environments, humans, animals, and both photorealistic and imaginative content. For each image (3D) or video (4D) we synthesize six camera trajectories, thereby yielding 354,186 videos for the 3D setting and 44,268 videos for the 4D setting.

Baselines. We benchmark our method against the state-of-the-art approaches to generative single-image-to-3D reconstruction—namely, ZeroNVS (Sargent et al., 2024), ViewCrafter (Yu et al., 2024b), Wonderland (Liang et al., 2025a), and Bolt3D (Szymanowicz et al., 2025b). We note that, since *no source code is available* for the evaluation of these methods on our out-of-distribution evaluation set, we rely predominantly upon the quantitative comparisons reported in the respective papers.

Evaluation. In accordance with prior work, we evaluate our model on the task of single-image-to-3D using RealEstate10K (Zhou et al., 2018), DL3DV (Ling et al., 2024b), and Tanks and Temples (Knapitsch et al., 2017), adhering throughout to the evaluation protocol delineated in Wonderland (Liang et al., 2025a) and Bolt3D (Szymanowicz et al., 2025b). Performance is quantified by means of the standard reconstruction metrics—PSNR, SSIM, and LPIPS.



Figure 6: **Image-to-3DGS Generation.** We visualize five distinct views drawn from the generated 3DGS scenes.



Figure 7: **Ablations.** We visualize the ablation results by rendering an identical extreme novel viewpoint subsequent to image-to-3DGS generation; the accompanying depth visualizations and ablations are furnished in Appendix C.4.

6.2 MAIN RESULTS

Quantitative results. The quantitative evaluations for single-image-to-3D are collated in Tab. 1, wherein our method surpasses all prior works on every benchmark and across every metric. It follows that the principal avenue for further elevating quality lies in the development of stronger video generative models, from which our reconstructions stand to benefit directly. Supplementary quantitative comparisons on the *Lyra* dataset are furnished in Appendix C.

Qualitative results. Novel-view renderings from our generated 3DGS scenes are visualized in Fig. 1 and Fig. 6; the corresponding video results will be released upon acceptance. Our method generates high-quality novel-view content for both 3D and 4D scenes throughout unseen regions, all the while preserving fidelity to the input image or video. Qualitative comparisons on the *Lyra* dataset are likewise deferred to Appendix C.

6.3 ABLATIONS

In Tab. 2 we substantiate our design choices by ablating the salient components on a diverse set of out-of-distribution prompts, comparing throughout the 3DGS renderings against the videos generated by our camera-controlled video diffusion model.

Real data only. Forgoing self-distillation entirely, we train a model on real multi-view datasets alone—namely, RealEstate10K (Zhou et al., 2018) and DL3DV (Ling et al., 2024b). Although these datasets enjoy widespread use in prior work, they engender limited generalizability to out-of-distribution scenes.

Self-distillation + real data. Somewhat counterintuitively, joint training on self-distillation together with real data confers no improvement over self-distillation in isolation. This observation corroborates that self-distillation is of itself

Table 2: **Ablation study conducted on the *Lyra* dataset.**

Method	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$
Ours	24.77	0.837	0.224
Data			
real data only	19.08	0.659	0.413
self-distill. + real data	24.74	0.823	0.236
Loss			

sufficiently diverse and consistent to induce a competent reconstruction model.

No depth loss. The depth loss forestalls flat geometry and, indeed, marginally improves the image-based metrics; the corresponding depth maps are visualized in Appendix C.4.

No opacity pruning. Here we retain the entirety of the Gaussians rather than pruning them by opacity. Learning to prune the preponderance of the 3D Gaussians renders the output representation more compact and marginally elevates visual quality; rendering at $H = 704$ and $W = 1280$ consumes 18ms with pruning as against 30ms without, a $1.67\times$ acceleration.

No LPIPS loss. Incorporating the LPIPS loss augments robustness to input inconsistencies and heightens the preservation of high-frequency detail.

No multi-view fusion. A naive implementation would generate 3D Gaussians for each camera trajectory independently and thereafter amalgamate them into a single point cloud. We observe, however, pronounced advantages in learning the multi-view fusion across the disparate trajectories through our reconstruction blocks, within which each token attends to the remainder.

No Mamba-2. Substituting Transformer-only blocks for our joint Transformer/Mamba-2 configuration yields marginally inferior quality. A single forward pass with $V = 6$, $L = 121$, $H = 704$, and $W = 1280$ requires 3213ms with the joint blocks, in contradistinction to 20922ms with the Transformer-only variant—a $6.5\times$ acceleration.

No latent-based 3DGS. Prior methods operating in pixel space, such as BTimer (Liang et al., 2025b), are circumscribed to 12 input frames, whereas our approach accommodates up to 726. Operating the 3DGS decoder in pixel space rather than latent space therefore precipitates an out-of-memory failure.

7 CONCLUSION

In this work we have presented *Lyra*, a novel framework for 3D and 4D generation that is predicated upon nothing more than a single image or video input. Rather than assembling multi-view datasets, we enlist camera-controlled video diffusion models as teachers within a self-distillation framework governing a student 3DGS decoder. This decoder operates within the latent space of the video model and reconstructs 3D Gaussians directly, without any post-processing or optimization whatsoever—a design that markedly enhances generalizability and coverage across heterogeneous scenes, while our dynamic extension attests to the feasibility of 4D generation from monocular video. At present, the scale and consistency of our generated scenes are circumscribed by the capacity of the underlying camera-controlled video diffusion model; strengthening the latter will accordingly unlock large-scale, consistent scene synthesis. A further compelling avenue for future work lies in investigating the assimilation of auto-regressive techniques (Chen et al., 2024a) into our framework so as to facilitate large-scale generation. Finally, the explicit modeling of motion and tracking information within the reconstruction network, in the spirit of concurrent work (Lin et al., 2025), holds the promise of improved visual motion quality.

8 ETHICS STATEMENT

In common with all generative AI technologies, our method is attended by risks of misuse, among them the production of misleading or synthetic 3D/4D content. Notwithstanding the existence of such risks, the principal aim of this research is to advance the fields of simulation, robotics, and embodied AI by furnishing scalable and controllable instruments for data generation. In this respect, we anticipate that our most consequential contributions will accrue through the following capabilities and insights.

- **The capacity to enable realistic simulation for embodied agents:** We supply controllable and physically consistent camera motion within generated 3D and 4D environments, thereby underpinning the training and evaluation of agents that must perceive, navigate, and interact with complex scenes.
- **Improved scalability for data engines:** By obviating both multi-view capture and expensive per-scene optimization, our method facilitates the production of high-fidelity, customizable scenes at scale—a capability particularly germane to research in robotics, reinforcement learning, and closed-loop simulation.
- **Deeper technical understanding:** As an academic study, this work furnishes a foundational case study of how camera control interacts with spatiotemporal generative models, to the benefit of research across computer vision, graphics, and embodied AI.

While we caution emphatically against the deceptive or wrongful deployment of this technology—whereby misleading or unverifiable media might be fabricated from realistic generative outputs—we hold that the adoption of appropriate safeguards, such as provenance tracking, dataset documentation, and sound evaluation practices, will prove instrumental in ensuring that generative models advance science and society to salutary ends.

9 REPRODUCIBILITY STATEMENT

So as to permit the full reproduction of our results and to expedite future research in this area, our training and inference code, model weights, and data for both 3D and 4D generation will be released upon acceptance.

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A ADDITIONAL DETAILS OF VIDEO DIFFUSION MODEL

This section furnishes additional detail concerning the GEN3C (Ren et al., 2025) video diffusion model backbone introduced in Sec. 3.1, together with the means by which we further augment its 3D consistency.

Conservative mask refinement. A cardinal limitation of GEN3C’s forward warping is background leakage, whereby occluded object parts in the source view (e.g., the lion’s body in Fig. 8) come to be visible in novel viewpoints yet are not duly registered within the disocclusion mask. The upshot is an incomplete foreground completion on the part of the video diffusion model, as exhibited in Fig. 8 (second and fourth columns).

To redress this, we adopt a conservative strategy: inasmuch as the geometry of the occluded regions in $\mathbf{P}^{t,v}$ is unknown, we presume all such areas to be occupied and derive refined disocclusion masks accordingly. This expedient enables the video model to reason about potentially visible foreground regions that standard point-based rendering had overlooked.

Our approach constructs a triangular mesh from the camera-space points $\mathbf{P}_c^{t,v} = \mathbf{C}^t \mathbf{P}^{t,v}$ by connecting spatially adjacent pixels, in a manner analogous to Hu et al. (2021):

$$\mathcal{M} = \bigcup_{(u,v)} \{ \triangle(\mathbf{p}_{u,v}, \mathbf{p}_{u+1,v}, \mathbf{p}_{u,v+1}), \triangle(\mathbf{p}_{u+1,v}, \mathbf{p}_{u+1,v+1}, \mathbf{p}_{u,v+1}) \} \quad (2)$$

where $\mathbf{p}_{u,v}$ denotes the 3D point at pixel (u, v) in $\mathbf{P}_c^{t,v}$. This surface mesh functions as a boundary demarcating the visible from the invisible regions. For each rendered pixel we juxtapose the standard point-based depth $\mathbf{D}^{t,v}$ with the mesh-interpolated depth $\mathbf{D}_{\mathcal{M}}^{t,v}$ obtained through ray–surface intersection:

$$\mathbf{M}^{t,v}(u, v) = \begin{cases} 0 & \text{if } \mathbf{D}_{\mathcal{M}}^{t,v}(u, v) < \mathbf{D}^{t,v}(u, v) - \epsilon \\ \mathbf{M}_{\text{orig}}^{t,v}(u, v) & \text{otherwise} \end{cases}, \quad (3)$$

where $\mathbf{M}_{\text{orig}}^{t,v}$ denotes the original disocclusion mask arising from forward warping and ϵ is a small tolerance factor. Whenever the mesh surface lies closer than the original points, we conservatively mask out these regions as prospective foreground disocclusions—a procedure that yields more dependable guidance for video generation, particularly under pronounced viewpoint changes, as exhibited in Fig. 8 (third and fifth columns).

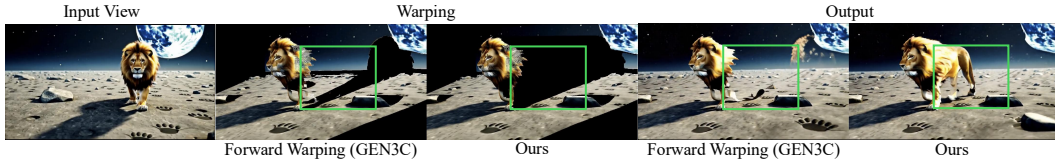


Figure 8: **Rendering function comparison.** Our model is founded upon GEN3C (Ren et al., 2025), which employs forward-warped images as camera-control conditioning. We juxtapose forward warping (left) against our conservative rendering function (right): the highlighted region, erroneously populated with background pixels under forward warping, is properly masked under ours, thereby enabling correct completion.

B MODEL DETAILS

Progressive training. Because training the 3DGS decoder directly on 6 trajectories of 121 frames at 704×1280 is prohibitively expensive, we instead train our network progressively, as tabulated in Tab. 3, and we further employ the final static pre-trained model to initialize its dynamic counterpart. The total training consumes 6 days on 8 NVIDIA A100 80 GB GPUs, with gsplat (Ye et al., 2025) serving as the 3DGS implementation.

Subsampling Gaussians. The generation of pixel-aligned 3D Gaussians scales linearly with each spatial and temporal resolution. Since our generated scenes represent 726 frames at 704×1280 resolution, a naive treatment would entail 654,213,120 per-pixel Gaussians. We therefore generate but

Table 3: **Progressive training regimen.**

Stage	$H \times W$	L	V	S	B	Steps
Static						
1	176×320	17	1	17	4	10k
2	176×320	49	1	49	4	2.5k
3	352×640	49	1	49	2	2.5k
4	704×1280	49	1	49	1	2.5k
5	704×1280	121	1	9	1	57.5k
6	704×1280	121	1–6	9	1	7k
Dynamic						
7	704×1280	121	6	12	1	10k

a single 3D Gaussian per 8×8 spatial neighborhood, thereby diminishing the total by a factor of 64, to 10,222,080; our opacity-based pruning attenuates this figure still further to 2,044,416, rendering the output appreciably more compact.

Joint Transformer-Mamba blocks. We adopt the reconstruction block design introduced in Long-LRM (Ziwen et al., 2024), wherein a single block comprises one Transformer layer succeeded by seven Mamba-2 layers; repeating this block twice yields 16 layers with 512 hidden dimensions. It bears noting that our network is substantially smaller than those of Long-LRM (Ziwen et al., 2024) or Wonderland (Liang et al., 2025a), both of which deploy 24 layers with 1024 hidden dimensions. Although pure Transformers are, as a rule, more expressive, they prove markedly slower in both training and inference than Mamba-2. Since we trained the two configurations within an identical training budget, we conjecture that Transformer-only blocks would demand considerably more training iterations before attaining parity in rendering quality.

Camera encoding. For camera encoding we employ Plücker embeddings, in keeping with prior work (Kant et al., 2024; Zhang et al., 2024e). These embeddings are computed at full spatial and temporal resolution and thereafter projected into the video latent space by means of the pre-trained RGB encoder. Since the RGB encoder anticipates 3 input channels whereas Plücker embeddings are 6-dimensional, we separately encode the 3-dimensional ray directions and the 3-dimensional cross product of the ray directions and origins, and concatenate the resulting latent encodings along the channel dimension.

Time encoding. In analogous fashion, to render the 1-dimensional time input compatible with the 3-channel input expected by the RGB encoder, we concatenate a 2-dimensional sinusoidal embedding to the original time value along the channel dimension, and replicate these values across the spatial dimensions prior to encoding the time into the video latent space. This procedure permits the independent encoding of both the source and target times into the compressed latent representation.

Dynamic data augmentation. We visualize our dynamic data augmentation procedure for two representative camera trajectories in Fig. 9. The procedure engenders pairs of supervision views that apprehend the scene from an identical motion state yet from divergent extreme viewpoints.

C ADDITIONAL EVALUATION

C.1 ADDITIONAL BASELINE DETAILS

We additionally benchmark our model against BTimer (Liang et al., 2025b), a recent approach to joint 3D and 4D reconstruction from posed images or videos. We approached the authors of BTimer, who conducted experiments on our behalf using the same scenes as *Lyra*. Since BTimer is purely regression-based, we integrate it with our GEN3C (Ren et al., 2025) video diffusion backbone: concretely, we employ the same camera trajectories as our model and supply the RGB-decoded videos from GEN3C as input to BTimer. BTimer operates in pixel space and requires 12 input images; as discussed in the main paper, however, pixel-space models such as BTimer exhaust available memory when confronted directly with the 726 high-resolution frames generated by GEN3C. To circumvent this, we uniformly subsample the GEN3C frames: for static evaluations we select 12 frames that evenly canvass the available viewpoints, whereas for dynamic evaluations we sample 11 frames spanning the viewpoint range and additionally incorporate the bullet-time frame corresponding to the

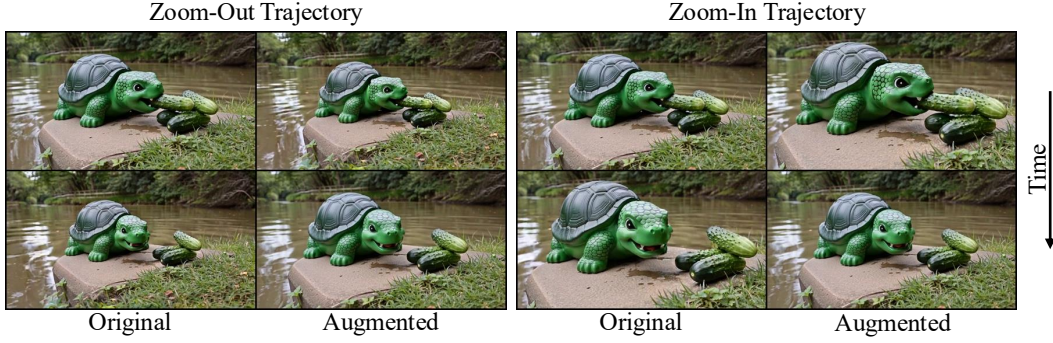


Figure 9: **Dynamic data augmentation videos.** We augment the supervision data with a motion-reversed video, thereby ensuring that each timestep is observed under full spatial coverage and, in consequence, averting low-opacity artifacts at the early timesteps. We present two representative trajectories—namely, zoom-out and zoom-in—and visualize their corresponding augmented videos, in which the camera motion is reversed.



Figure 10: **Image-to-3DGS Comparison.** We compare our method with BTimer (GEN3C), visualizing five views from the generated scenes, and discern markedly fewer artifacts together with appreciably higher fidelity.

motion under reconstruction, thereby ensuring that the target motion state is invariably present as a reference. We denote this integrated baseline BTimer (GEN3C), i.e., BTimer in combination with GEN3C.

C.2 ADDITIONAL STATIC 3D EVALUATION

For the task of image-to-3DGS generation, we present qualitative comparisons with BTimer (GEN3C) in Fig. 10 and the corresponding quantitative results in Tab. 4a, from which we discern both superior quality and a diminished incidence of artifacts on the part of our method.

C.3 DYNAMIC 3D EVALUATION

BTimer (GEN3C) comparison. For dynamic 3D evaluation, we compare our method with BTimer (GEN3C), as elaborated in Sec. C.1, on 100 out-of-distribution videos drawn from our dynamic *Lyra* dataset. In the interest of a fair comparison, we crop the video to 512×512 , since

Table 4: Quantitative results across the static and dynamic *Lyra* datasets.

Method	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	Method	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$
BTimer (GEN3C)	16.32	0.580	0.427	BTimer (GEN3C)	20.29	0.687	0.315
Ours	24.92	0.834	0.183	Ours	23.07	0.779	0.231

(a) Comparison on the static *Lyra* dataset.

(b) Comparison on the dynamic *Lyra* dataset.

BTimer was trained predominantly at that resolution. The results, reported in Tab. 4b, reveal that our method substantially outperforms BTimer (GEN3C).

C.4 DEPTH VISUALIZATION

We visualize the depths recovered from our generated 3DGS in Fig. 11. The employment of depth supervision forestalls flat geometries without any concomitant sacrifice in visual quality.

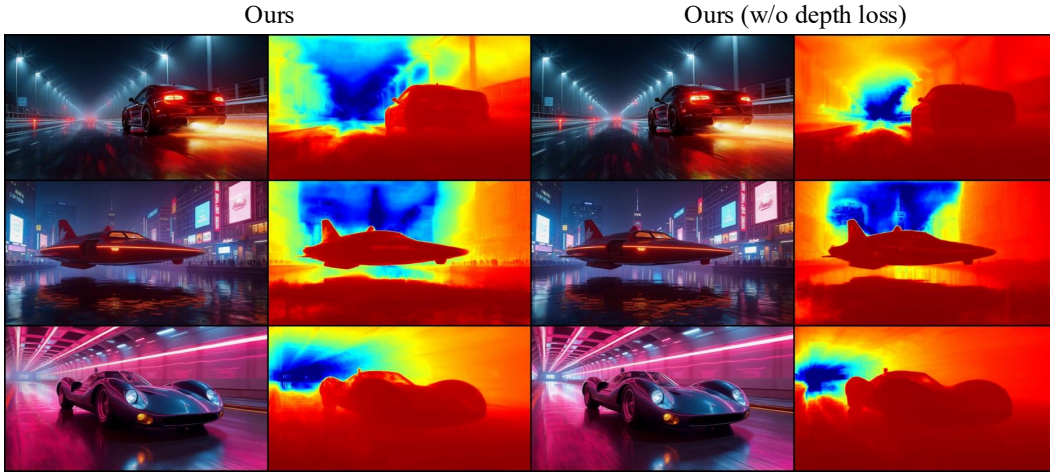


Figure 11: **Depth loss ablation.** We visualize the 3DGS renderings alongside their corresponding depth maps. Enlisting the depth loss as an additional supervisory signal forestalls flat geometries without any concomitant sacrifice in visual quality.

D ROBOT SIMULATION

A foremost motivation for constructing a diverse 3D generator is its capacity to synthesize simulation environments for autonomous agents. To this end, we develop an end-to-end pipeline: we first generate 3D scenes from text with our model, export the 3D Gaussians as a .ply file, and thereafter convert the .ply files into the .usdz format by means of 3DGUT (Wu et al., 2025a) and its export functionality. The resulting .usdz file may then be imported into the NVIDIA Isaac robot development platform as a physically based virtual environment, within which AI-based robots can be trained and evaluated under heterogeneous conditions. Preliminary demonstration results will be released upon acceptance; a screenshot of the interface is provided in Fig. 12.

E RELATED WORK

In this section we canvass additional related work, and we delineate the distinctions from the prior works CAT3D (Gao et al., 2024b) and Bolt3D (Szymanowicz et al., 2025b) in Fig. 13.

3D generation. Early approaches to 3D generation were directed chiefly at single object categories, extending GANs into the 3D domain by leveraging neural renderers as an inductive bias (DeVries et al., 2021; Chan et al., 2022; Or-El et al., 2022; Schwarz et al., 2022; Bahmani et al., 2023b). With the advent of CLIP-based supervision (Radford et al., 2021), the field progressed toward more flexible

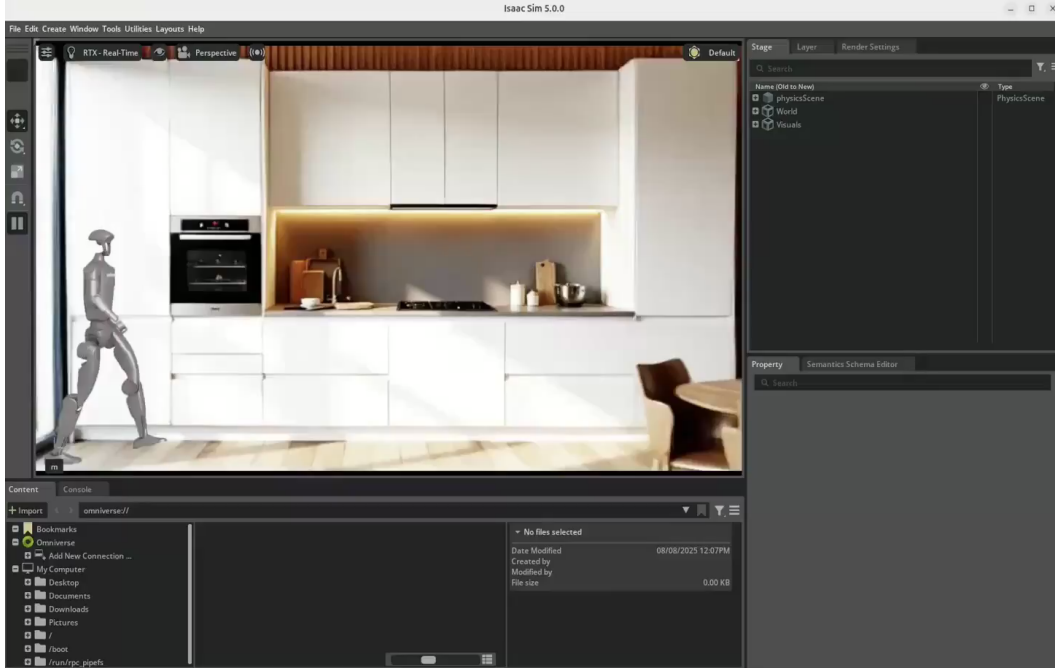


Figure 12: **Robot Simulation.** We visualize a single frame of a robot simulation conducted within the Isaac Sim 5.0 framework, which ingests a 3DGS scene generated by our method as its input.

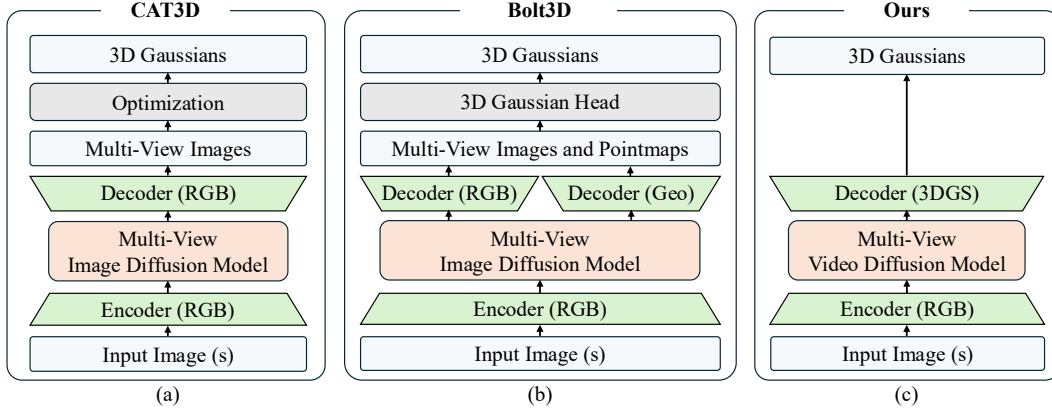


Figure 13: **Approaches for 3D Generation.** CAT3D (Gao et al., 2024b) proposed a multi-view image diffusion model that emits images from novel viewpoints, which are subsequently reconstructed into 3D Gaussians by optimization. Bolt3D (Szymanowicz et al., 2025b) fine-tunes CAT3D to output pointmaps by means of a geometry autoencoder and, in lieu of optimization, predicts the 3D scene with a feed-forward 3D Gaussian head. Our work, by contrast, is founded upon a pre-trained camera-controlled video diffusion model and *directly* decodes the multi-view video latents into 3D Gaussians.

and diverse asset creation, thereby enabling both text-driven generation and editing (Chen et al., 2018; Jain et al., 2022; Sanghi et al., 2022; Jetchev, 2021; Gao et al., 2023; Wang et al., 2022). More recently still, diffusion models have appreciably improved quality by supplanting CLIP guidance with Score Distillation Sampling (SDS) (Poole et al., 2023; Wang et al., 2023b; Lin et al., 2023a; Chen et al., 2023b; Liang et al., 2023; Wang et al., 2023a; Li et al., 2024e; He et al., 2024b; Ye et al., 2024; Liu et al., 2024b; Yu et al., 2023; Katzir et al., 2024; Lee et al., 2024a; Sun et al., 2024a).

With a view to further reinforcing structural coherence, several methods enforce multi-view consistency by generating scenes from disparate perspectives (Lin et al., 2023b; Liu et al., 2023; Shi et al., 2024; Feng et al., 2024a; Liu et al., 2024a; Kim et al., 2023; Voleti et al., 2024; Höllein et al., 2024;

Tang et al., 2024c; Gao et al., 2024b; Wang et al., 2025f; Kant et al., 2025; Yuan et al., 2025; Ren et al., 2024b), whereas others enlist iterative inpainting as a strategy for scene synthesis (Höllein et al., 2023; Shriram et al., 2024). A further line of inquiry explores the lifting of 2D images into 3D by means of NeRF (Mildenhall et al., 2020), 3D Gaussian Splatting (Kerbl et al., 2023), or mesh-based representations coupled with diffusion models (Chan et al., 2023; Tang et al., 2023a; Gu et al., 2023; Liu et al., 2024c; Yoo et al., 2023; Tewari et al., 2023; Qian et al., 2024b; Long et al., 2024; Wan et al., 2024; Szymanowicz et al., 2023; Lu et al., 2024).

Feed-forward 3D models. Complementary efforts investigate fast, feed-forward techniques that predict 3D content directly from image or text inputs (Hong et al., 2024; Li et al., 2024b; Xu et al., 2024e;d; Zhang et al., 2024a; Han et al., 2024; Jiang & Wang, 2024; Xie et al., 2024a; Tang et al., 2024b; Tochilkin et al., 2024; Qian et al., 2024a; Szymanowicz et al., 2024; 2025a; Liang et al., 2025a; Szymanowicz et al., 2025b; Schwarz et al., 2025; Yang et al., 2025; Zhang et al., 2025a). In contradistinction to our method, however, these approaches remain confined to the production of static 3D scenes, while others are restricted to specific subject matter such as faces (Kirschstein et al., 2025). Concurrent work (Liang et al., 2025b; Xu et al., 2025) does address real scenes, yet cannot accommodate diverse generated scenes or pronounced viewpoint changes.

4D generation. Recent years have witnessed rapid progress in 4D generation—that is, dynamic 3D scene synthesis—much of it guided by input text prompts or images. Following the introduction of large-scale generative models for this task (Singer et al., 2023), subsequent works have registered notable advances in both visual fidelity and motion quality (Ren et al., 2023; Ling et al., 2024a; Bahmani et al., 2024b; Zheng et al., 2024b; Gao et al., 2024a; Yang et al., 2024a; Jiang et al., 2024b; Miao et al., 2024; Li et al., 2024a; Zhang et al., 2024g; Yuan et al., 2024; Jiang et al., 2024a).

Although text conditioning enjoys widespread adoption, alternative methods endeavor to lift 2D images or videos into dynamic 3D scenes (Ren et al., 2023; Zhao et al., 2023; Yin et al., 2023; Pan et al., 2024; Zheng et al., 2024b; Ling et al., 2024a; Gao et al., 2024a; Zeng et al., 2024b; Chu et al., 2024; Wu et al., 2024; Yang et al., 2024b; Wang et al., 2024c; Feng et al., 2024b; Sun et al., 2024c; Zhang et al., 2024c; Ren et al., 2024a; Lee et al., 2024b; Li et al., 2024c; Van Hoorick et al., 2024; Uzolas et al., 2024; Chai et al., 2024; Liang et al., 2024; Li et al., 2024d; Xie et al., 2024b; Li et al., 2025a; Wang et al., 2025e; Zhu et al., 2025; Zhou et al., 2025; Hu et al., 2025; Park et al., 2025). Further efforts explore the incorporation of physics priors into the generation pipeline (Lin et al., 2024; Huang et al., 2024b; Zhang et al., 2024f; Kiray et al., 2025), or the enhancement of motion controllability through template-driven approaches (Zhang et al., 2024d; Sun et al., 2024b; Chen et al., 2024b), while a parallel direction accentuates compositional and interactive 4D generation (Bahmani et al., 2024a; Xu et al., 2024b; Cao et al., 2024; Yu et al., 2024a; Zeng et al., 2024a; Zhu et al., 2024).

Yet another strand of research extends 3D GAN frameworks into the 4D setting by training directly on 2D video data (Bahmani et al., 2023a; Xu et al., 2023). Such methods are, however, characteristically hampered by small-scale, single-category datasets that curtail generalization; moreover, the preponderance of existing approaches remain object-centric and frequently neglect background elements. In consequence, their overall visual fidelity trails the high photorealism attained by recent video generation models—precisely the fidelity that our method turns to its advantage.

Camera-conditioned video models. Recent progress has concentrated upon the incorporation of camera control into video diffusion models. The pioneering work MotionCtrl (Wang et al., 2024e) introduced camera conditioning by augmenting pre-trained video models (Chen et al., 2023a; Blattmann et al., 2023) with extrinsic matrices. Subsequent efforts (He et al., 2024a; Xu et al., 2024c; Kuang et al., 2024; Ju et al., 2025; He et al., 2025; Li et al., 2025c; Wang et al., 2025c; 2024d; Lei et al., 2025; Liu et al., 2025) refined this conditioning by representing cameras via Plücker coordinates, whereas others follow MotionCtrl in simply flattening the poses (Wang et al., 2025b). A distinct line of research (Hu et al., 2024; Xiao et al., 2024; Ling et al., 2024c; Hou et al., 2024) enabled camera-motion control without the introduction of any additional trainable parameters. For all these advances, however, the foregoing methods rely uniformly upon U-Net-based architectures; still others resort to low-rank formulations for camera control (Zhang et al., 2025b).

Whereas several recent works persist in extending U-Net-based approaches to camera control (Zhao et al., 2024; Xu et al., 2024a; Zheng et al., 2024a; Zhang et al., 2024b; Yu et al., 2024b; Lei et al., 2024), others have begun to explore diffusion transformers. Cheong et al. (2024), for instance, propose one such transformer-based framework, albeit one whose scene and visual quality remain limited, while

1566 DimensionX (Sun et al., 2024d) introduces joint space-time control within diffusion transformers
1567 yet remains reliant upon pre-defined, non-continuous camera trajectories. Other works (Huang
1568 et al., 2024a; Mai et al., 2025; Jiang et al., 2025b) instead pursue pose and geometry estimation
1569 with a video DiT, and CAT4D (Wu et al., 2025b) advances a multi-view video diffusion model
1570 derived by fine-tuning a multi-view backbone. Large-scale synthetic data generation confers a further
1571 boost in performance, as evidenced by SynCamMaster (Bai et al., 2025b) and ReCamMaster (Bai
1572 et al., 2025a), both of which attain strong results on camera-controlled video generation by way of
1573 transformer-based architectures. Finally, 4DiM (Watson et al., 2024) trains a space-time diffusion
1574 model from scratch for novel-view synthesis from a single input image. A separate line of work
1575 edits the camera movements of videos possessing underlying motion through camera retargeting (Seo
1576 et al., 2025; Ma et al., 2025). One especially prevalent direction has been to employ a spatio-temporal
1577 cache—constructed, for example, from point clouds or optical flow—to condition camera-controlled
1578 video generation (Wu et al., 2025c; Ren et al., 2025; Wang et al., 2025d; Cao et al., 2025; YU et al.,
1579 2025; Jin et al., 2025; Xing et al., 2025; Yan et al., 2025; Li et al., 2025b; Gu et al., 2025).